Linear equations in one variable — Part 1

Algebra · Math · 56 questions

1.1 85789084-6a50-4310-ae89-d2067332bac0 · Easy

If $4x - 28 = -24$, what is the value of $x - 7$?

A. -24

B. -22

C. -6

D. -1

1.2 a7ee5a49-de3b-4c5b-a46e-d593e051326d · Easy

If $x = 7$, what is the value of $x + 20$?

A. 13

B. 20

C. 27

D. 34

1.3 cb01072a-91f8-4bc2-b74f-51d996629f69 · Easy

If $4x = 3$, what is the value of $24x$?

A. $\frac{9}{2}$

B. 6

C. 18

D. 72

1.4 96677815-bb45-424b-a4e5-ec40dbe43bb8 · Easy

On the first day of a semester, a film club has 90 members. Each day after the first day of the semester, 10 new members join the film club. If no members leave the film club, how many total members will the film club have 4 days after the first day of the semester?

A. 400

B. 130

C. 94

D. 90

1.5 67336474-3a8f-4702-9694-98a3abdcc31a · Easy

If $\frac{x}{8} = 5$, what is the value of $\frac{8}{x}$?

Student-produced response: _____

1.6 62a3416c-5c61-4965-82b5-d0803a5d0182 · Easy

If $6 + x = 9$, what is the value of $18 + 3x$?

Student-produced response: _____

1.7 5b069676-b4f1-4f88-aa86-55183f3f0055 · Easy

A principal used a total of 25 flags that were either blue or yellow for field day. The principal used 20 blue flags. How many yellow flags were used?

A. 5

B. 20

C. 25

D. 30

1.8 d4e10666-e119-46af-8dba-a8d527aeca75 · Easy

3 more than 8 times a number x is equal to 83. Which equation represents this situation?

A. $(3)(8)x = 83$

B. $8x = 83 + 3$

C. $3x + 8 = 83$

D. $8x + 3 = 83$

1.9 26011899-7805-4655-b434-d6a03ef7d8de · Easy

If $2 + x = 60$, what is the value of $16 + 8x$?

Student-produced response: _____

1.10 fe2efe64-2759-4c9c-b6ac-4e4f9f1c9372 · Easy

What value of x is the solution to the equation $16x + 24 = 24x$?

A. -4

B. $\frac{3}{10}$

C. $\frac{1}{3}$

D. 3

1.11 022296-DC · Easy

If $3x + 2 = 8$, what is the value of $9x + 6$?

Student-produced response: _____

1.12 214d64ec-0e1a-4a3f-8fcb-73b3eecf4a6b · Easy

$$3x + 21 = 3x + k$$

In the given equation, k is a constant. The equation has infinitely many solutions. What is the value of k?

Student-produced response: _____

1.13 5b55a923-dce4-4bf9-a037-3f16263d1743 · Easy

If $9x + 4 = 67$, what is the value of $90x + 40$?

A. 7

B. 70

C. 130

D. 670

1.14 08682-DC · Easy

$$\frac{4x}{5} = 20$$

In the equation above, what is the value of x ?

A. 25

B. 24

C. 16

D. 15

1.15 9931da93-9c67-4834-ac2f-f3587e875e3e · Easy

If $x = 40$, what is the value of $x + 6$?

A. 34

B. 40

C. 46

D. 64

1.16 022240-DC · Easy

If $\frac{2n}{5} = 10$, what is the value of $2n - 1$?

A. 24

B. 49

C. 50

D. 99

1.17 8076db82-ac6c-4ac2-a6b7-46a9a4b4059b · Easy

If $3x - 8 = 7$, what is the value of $3x + 8$?

A. -1

B. 5

C. 13

D. 23

1.18 9d2c51d3-1801-4e1e-b2f2-9e58978c2f8e · Easy

The perimeter of an isosceles triangle is 36 feet. Each of the two congruent sides of the triangle has a length of 10 feet. What is the length, in feet, of the third side?

A. 10

B. 12

C. 16

D. 18

1.19 14894-DC · Easy

If $\frac{1}{2}x-\frac{1}{6}x=1$, what is the value of x ?

A. -4

B. $\frac{1}{3}$

C. 3

D. 6

1.20 8cb965d3-c0cb-4549-8b51-5decc372d910 · Easy

A total of 165 people contributed to a charity event as either a donor or a volunteer. 130 people contributed as a donor. How many people contributed as a volunteer?

A. 35

B. 130

C. 165

D. 330

1.21 d08114ae-ef2f-4e5b-a0ef-67d3bb996aa9 · Easy

If $3x = 30$, what is the value of $3x - 12$?

A. -2

B. 18

C. 22

D. 42

1.22 06902-DC · Easy

A librarian has 43 books to distribute to a group of children. If he gives each child 2 books, he will have 7 books left over. How many children are in the group?

A. 15

B. 18

C. 25

D. 29

1.23 84445f9f-c91d-47ad-a7fb-f1adeb091390 · Easy

If $8x = 6$, what is the value of $72x$?

A. 3

B. 15

C. 54

D. 57

1.24 bcd38cac-e6c1-411f-ae0b-50e4ae48adc1 · Easy

$$(p + 3) + 8 = 10$$

What value of p is the solution to the given equation?

A. -1

B. 5

C. 15

D. 21

1.25 8a9230ef-2284-4f11-9f4e-abff1fd55a4e · Easy

$$2.6 + x = 2.8$$

What value of x is the solution to the given equation?

Student-produced response: _____

1.26 7e8a4250-4838-4f9d-9e6c-24235e682f46 · Easy

A manager is responsible for ordering supplies for a shaved ice shop. The shop's inventory starts with 4,500 paper cups, and the manager estimates that 70 of these paper cups are used each day. Based on this estimate, in how many days will the supply of paper cups reach 1,700?

A. 20

B. 40

C. 60

D. 80

1.27 974b35b6-beab-44b9-8d36-7f18c4daa0d1 · Easy

$$3x + 5(x + 4) = 76$$

What value of x is the solution to the given equation?

A. 7

B. 8

C. 56

D. 72

1.28 0f1aa9d1-12d4-4e22-b2aa-c753912a2a2e · Easy

$$8x = 88$$

What value of x is the solution to the given equation?

A. 11

B. 80

C. 96

D. 704

1.29 c754edce-0af7-420c-8fc5-0e5f0cd80b73 · Easy

A line segment that has a length of 115 centimeters (cm) is divided into three parts. One part is 47 cm long. The other two parts have lengths that are equal to each other. What is the length, in cm, of one of the other two parts of equal length?

Student-produced response: _____

1.30 9055c207-d1fc-4110-be3c-1ea08fcd52ab · Easy

If $6n = 12$, what is the value of $n + 4$?

Student-produced response: _____

1.31 c57e37c9-32b5-4879-b56c-ae8aec1a46fc · Easy

What value of p satisfies the equation $5p + 180 = 250$?

A. 14

B. 65

C. 86

D. 250

1.32 0aa2936b-88d9-4e8d-a776-486d5d1f93ba · Easy

$$4x + 5 = 165$$

What is the solution to the given equation?

Student-produced response: _____

1.33 c33a8ce0-debc-4709-b0d7-3f1800613186 · Easy

$$10x = 86$$

What value of x is the solution to the given equation?

Student-produced response: _____

1.34 020349-DC · Easy

$10 = 2x + 4$

How many solutions exist to the equation shown above?

A. None

B. Exactly 1

C. Exactly 3

D. Infinitely many

1.35 7ea2442b-48e4-447b-a75d-c1614b613123 · Easy

If $3x - 27 = 24$, what is the value of $x - 9$?

A. 1

B. 8

C. 24

D. 35

1.36 ee16e929-f3b9-4883-9cb5-c95afc6fb0df · Easy

For a restaurant owner to purchase a pizza oven at a total price of $13{,}000$ dollars, a onetime down payment is required, and then fixed monthly payments are made for the remaining amount owed for the pizza oven. The equation $13{,}000 = 2{,}800 + 200t$ represents this situation, where t is the number of fixed monthly payments that are made. Which of the following is the best interpretation of 200 in this context?

A. The amount, in dollars, of the down payment

B. The amount, in dollars, of each fixed monthly payment

C. The total amount, in dollars, paid for the pizza oven after t fixed monthly payments

D. The total number of fixed monthly payments

1.37 2ed3a51e-14c9-41a5-8246-b0df7d098ee8 · Easy

The perimeter of an isosceles triangle is 83 inches. Each of the two congruent sides of the triangle has a length of 24 inches. What is the length, in inches, of the third side?

Student-produced response: _____

1.38 05768-DC · Easy

Which of the following is equivalent to $4x+6=12$?

A. $2x+4=6$

B. $x+3=3$

C. $3x+2=4$

D. $2x+3=6$

1.39 ab4c9e76-1929-4d3f-ad64-4a63c8386545 · Easy

If $5x = 20$, what is the value of $15x$?

A. 7

B. 12

C. 23

D. 60

1.40 07141-DC · Easy

One pound of grapes costs $2. At this rate, how many dollars will c pounds of grapes cost?

A. $2c$

B. $2+c$

C. $\frac{2}{c}$

D. $\frac{c}{2}$

1.41 208ae094-8f06-4329-9993-51169b92549f · Easy

$$w + 7 = 357$$

What value of w is the solution to the given equation?

A. 51

B. 350

C. 364

D. 3,577

1.42 2870e625-1e5c-4b4d-be09-4cc08fc8d83b · Easy

John paid a total of $\$ 165$ for a microscope by making a down payment of $\$ 37$ plus p monthly payments of $\$ 16$ each. Which of the following equations represents this situation?

A. $16p - 37 = 165$

B. $37p - 16 = 165$

C. $16p + 37 = 165$

D. $37p + 16 = 165$

1.43 b80105ad-11a3-443a-9dc8-51472ba42bbf · Easy

Nasir bought 9 storage bins that were each the same price. He used a coupon for $\$63$ off the entire purchase. The cost for the entire purchase after using the coupon was $\$27$. What was the original price, in dollars, for 1 storage bin?

Student-produced response: _____

1.44 87d05c33-deec-41ec-8d9c-f0a0094beae6 · Easy

A gym charges its members a onetime $\$ 36$ enrollment fee and a membership fee of $\$ 19$ per month. If there are no charges other than the enrollment fee and the membership fee, after how many months will a member have been charged a total of $\$ 188$ at the gym?

A. 4

B. 5

C. 8

D. 10

1.45 70f8f5ea-a016-4c7d-ad77-2b55a0b1c990 · Easy

What value of p satisfies the equation $2p + 275 = 325$?

A. 5

B. 25

C. 48

D. 300

1.46 f815b832-89ea-4f49-bc05-5f5c25cdb59f · Easy

$$x + 40 = 95$$

What value of x is the solution to the given equation?

Student-produced response: _____

1.47 015193-DC · Easy

What is the solution to the equation $2x + 3 = 7$?

A. 1

B. 1.5

C. 2

D. 4

1.48 1da5e7f6-df74-4a25-9976-1f105af7942e · Easy

If $7x = 28$, what is the value of $8x$?

A. 21

B. 32

C. 168

D. 224

1.49 9df9861f-e14c-45b9-be5c-1d9bb6fbafba · Easy

If $2x = 12$, what is the value of $9x$?

Student-produced response: _____

1.50 f2ee68d6-91e0-4510-a278-ad0aa7e1153c · Easy

$$72x - 3 = 63$$

Which equation has the same solution as the given equation?

A. $2x - 3 = 9$

B. $2x - 3 = 56$

C. $2x - 21 = 63$

D. $2x - 21 = 70$

1.51 cab807ab-67ed-464f-a4b2-ece6edb417b6 · Easy

$$13x = 112 - x$$

What value of x is the solution to the given equation?

Student-produced response: _____

1.52 5bbff16a-8cdc-4d4c-b45c-545cf12538e2 · Easy

$$k + 12 = 336$$

What is the solution to the given equation?

A. 28

B. 324

C. 348

D. 4,032

1.53 151077e0-2b11-4c07-bc61-40cb9fc124a8 · Easy

Henry receives a $\$ 60.00$ gift card to pay for movies online. He uses his gift card to buy 3 movies for $\$ 7.50$ each. If he spends the rest of his gift card balance on renting movies for $\$ 1.50$ each, how many movies can Henry rent?

A. 10

B. 25

C. 35

D. 40

1.54 10931-DC · Easy

For what value of w does $w-10=2(w+5)$?

A. 5

B. 0

C. -15

D. -20

1.55 5839b7ae-3235-44a9-9666-5735f17e34d8 · Easy

$$4x + 1 = 33$$

Which equation has the same solution as the given equation?

A. $4x = 32$

B. $4x = 5$

C. $4x = 1$

D. $4x = -32$

1.56 a3e37333-8679-4d42-b7c8-1b15541bd74d · Easy

$$16x + 30 = 190$$

Which equation has the same solution as the given equation?

A. $16x = 30$

B. $16x = 130$

C. $16x = 160$

D. $16x = 190$
